

Coordinator Survey Console

HighTowers-Web - Wing planning aid for Low Level Route tower surveys

Audience: Wing coordinators, squadron leadership, and aircrew who need to understand how staffing decisions are made.

What this tool is

- How many sorties does each team need?
- Which contiguous MTR waypoint ranges should each aircraft fly?
- How do 1 vs 2 vs 3 teams compare on total wing sorties?

It uses NASR corridor width, your flight-plan waypoint sequence, departure airports you choose, and a per-sortie distance budget (typically 400-500 NM). It models G1000 parallel-track offsets on each side of the centerline.

It is not a weather tool, MOA planner, ATC product, or ForeFlight replacement. Crews still fly corridors in ForeFlight Military Flight Bag.

Who uses what

Wing coordinator	Coordinator Survey Console - staffing what-if, sortie briefs
Aircrew	Main app - flight plans, tower analysis, maps, PDF export, content packs
Wing administrator	Content Pack console - pack lifecycle (separate from coordinator planning)

How coordinators work in practice

1. Start with available squadrons - CAP teams operate from airports near their home base. The tool does not pick airports for you.
2. Build or open a flight plan for the MTR; open the console from Flight Plans.
3. Run a what-if - enter Team 2 and/or Team 3 airports; choose a mode or Compare 1 vs 2 / Compare 2 vs 3.
4. Brief teams - sortie rows: waypoint from-to, offsets, ferry and along-route NM.
5. Re-run when the roster changes - weather or availability may drop a team the day before: recalculate with fewer teams, or swap in another squadron's airport and re-brief everyone.

Management one-liner: The coordinator names which teams and airports can play; HighTowers calculates how to divide the route and sorties to cover the MTR with the fewest flights practical under the distance cap-and you can re-run when the roster changes.

Planner modes

1 team	One aircraft: inner then outer passes on the full route from one base.
2 teams	Parallel: opposite sides of the corridor; same geography, different bases.
3 teams	Geographic split: three segments; each team flies both sides of its segment. Optimizer picks splits and base assignments.
Compare 1 vs 2	One aircraft (full corridor) vs two aircraft (opposite sides).
Compare 2 vs 3	Two aircraft (opposite sides) vs three aircraft (geographic split).

What the optimizer does (and does not)

Does	Minimize total wing sorties (tie-break NM); pack waypoint chains into sorties under the budget; include ferry and offset legs; optimize each sortie's start end; for 3-team mode, search segment boundaries and team assignments.
Does not	Choose squadrons or airports; change the MTR waypoint list; replace ForeFlight or weather judgment.

Reading the results

Inner / Outer	Left and right of centerline (asymmetric spans use different offset lists).
Offsets	G1000 parallel-track spacing in NM (default: 3, 9, 15, outer margin).
Ferry in / out	Base to/from route entry or exit for that sortie.
Along route	Directed offset legs on the assigned waypoint chain.
Warning in Total	Over NM budget - shorter segment or more teams.

URLs (NM Wing pilot hosting)

Crew app	https://gvdurfee.github.io/HighTowers-Web/
Coordinator console	Flight Plans -> Coordinator Survey Console, or /coordinator/survey?plan=<id>
Wing admin	/admin/content-packs - wing staff only

After updates on main, hard refresh (Cmd+Shift+R). Pages rebuilds automatically; Railway API updates when server/ changes.

Wing planning aid only. Verify corridors and procedures in ForeFlight Military Flight Bag before flying.

Source: docs/COORDINATOR_SURVEY_CONSOLE_HANDOUT.md